

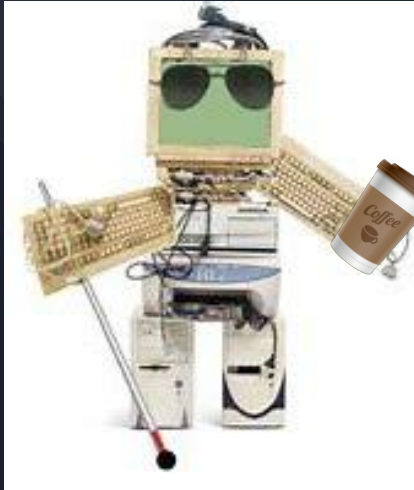
Assisting Blind people to Orient and Commute:

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Course: COGS 100 (*winter '23*) - *Cyborgs, now, and in the future*

Professor: Dr. Simpkins

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Route:

UCSD Central Campus Trolley Station → Catalyst Building

10 Challenges faced by blind individuals:

1. **To find a way down the trolley**
2. **To find the direction to the destination**
3. Changing of elevation
4. **People**
5. **Cars**
6. Avoiding obstacles, and finding a clear path to the destination
7. **Maintaining directional orientation**
8. Unpredictable factor: weather
9. **Crossing unregulated intersections**
10. Keeping in time

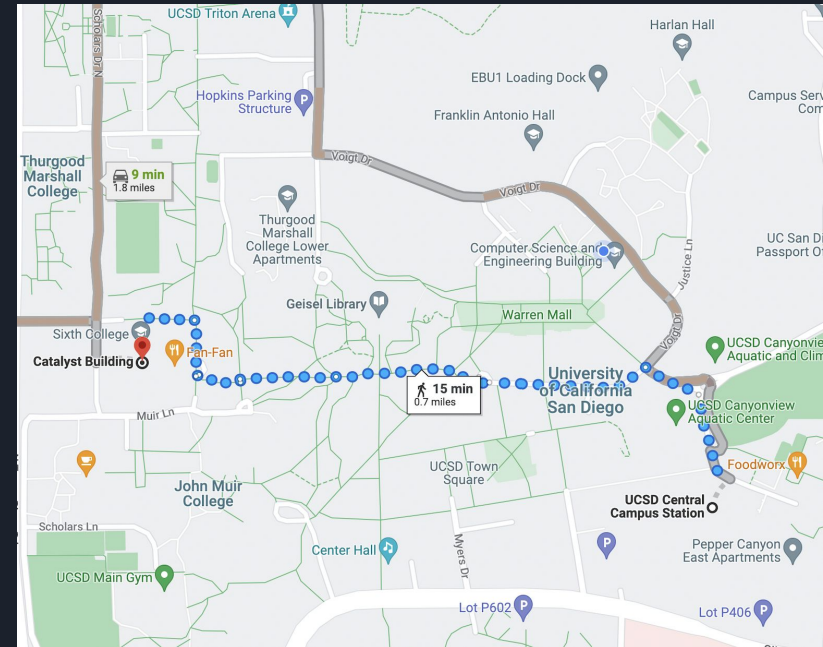


Fig. 1

The overall route from the trolley station to the lecture hall

Challenge #1: Getting off Trolley Station

- Locating a way down (Dückelmann et al. 2017)
- Stairs are difficult to traverse without vision
- Locating nearby elevator



Fig. 2
Entering the elevator/stairs
area from the trolley



Video. 1
Finding the correct button
on the elevator



Challenge #2: Direction

Nearly impossible without our device

- Knowing the layout/direction towards the end point
- Making sure you know the overall direction you're heading



Fig. 3

The subject does not know the way to go

Challenge #3: People/crowd

- Not running into others (Jeamwatthanachai et al. 2019)
- Knowing when to move out of people's way
- Gaging distance between moving bunches of people
- Being able to use audio feedback to locate surrounding people



Fig. 4
The crowd on campus

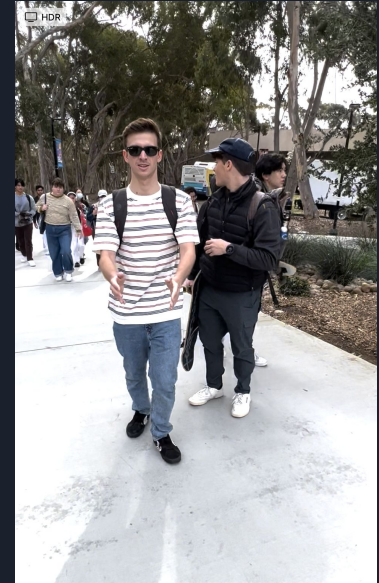


Fig. 5
The guide trying to help
the walking subject avoid
the individuals

Challenge #4: Cars

- Knowing when a car is approaching (can't gage how far away it is)



Fig. 6

The subject approaching a moving car without being notified

Challenge #5: Maintaining directional orientation

- Staying on the path to location
- Knowing when to alter course
- Understanding when drifting off direction
- Maintaining correct posture due to head feeling abnormally heavy



Video. 2

The subject trying to go straight but losing his sense of orientation unknowingly, and making a sharp turn under guidance

Challenge #6: Crossing unregulated intersections

- If there's no indication, it's hard to know when approaching the intersection
- Being able to see or hear approaching vehicle/bikes/scooters

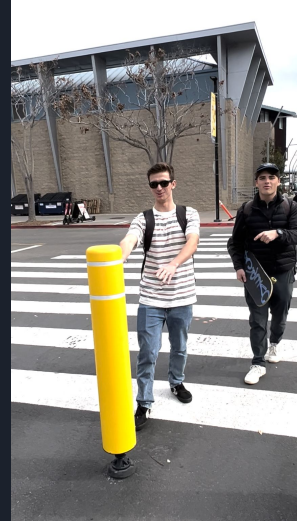


Fig. 7
The subject being extremely cautious when crossing the intersection while not knowing what the situation was

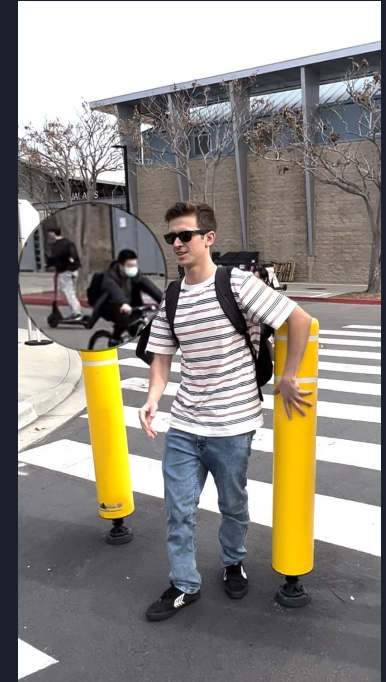
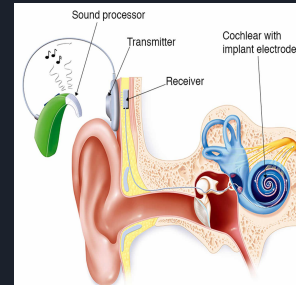


Fig. 8
The bikes and scooters moving around freely at intersections with no regulation

Current Strategies to Overcome challenges:

- White cane: To foresee obstacles or different surface elevations (bumps, curbs, etc)
(Y. Shiizu et al. 2007)
- Memory (looking ahead of time and having an understanding of the layout/map of the location the individual is heading to)
(Vecchi et al. 2010)
- Guide dogs: Can help with crossing busy intersections
(Whitmarsh, 2005)
- Cochlear implants (allows sense of sound)
(F. -G. Zeng et al. 2008)



Figs. 9-12
The current strategies being used

Prototype Sketch

- We will utilize the cochlear implant and bolster the design according to the challenges we faced.
- The implant will be placed on the users ear. It will consist of one sensor implemented to coincide with the GPS chip that will detect a certain radius (Morad, 2021.)
- The radius will be used to sense and prevent the user from colliding with objects within the radius (possibly radar system, (V. Di Mattia *et al.* 2016).
- The chip will then communicate with the cochlear implant via audio which will be heard by the user. (Nada *et al.* 2015)
- With a full charge the user can expect to use the device for a full 19 hours similar to that of the current cochlear implant.

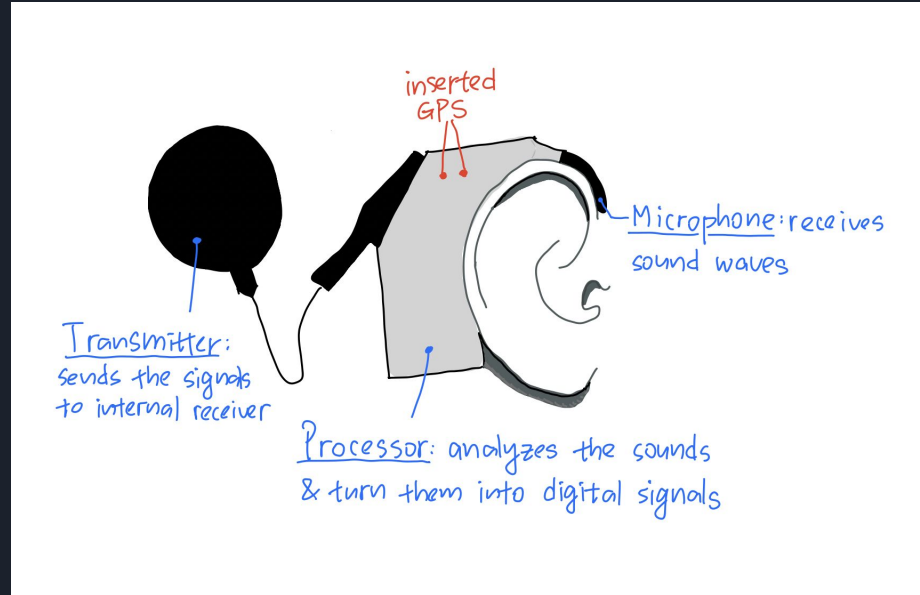


Fig. 13
Sketch 1: enhanced cochlear implant

Enhanced Strategies

- Implement GPS chip to enhance cochlear implant in order to deliver information regarding the route and radius to the user.
- Radius technology will be implemented to communicate with the GPS.
- The user will be able to plug in the desired destination of choice to the GPS navigation system.
- The GPS will notify the user when they steer away from the designated path to help combat the issues of direction (Bohle, P.R. 2019)
- Additionally, the GPS will notify the user when they are nearing an intersection.
- There will be two fields of radius. One (r) will be close to the user and much more sensitive to large and small objects such as humans, bushes, twigs etc. The larger radius (R) furthest from the user will be used to detect large objects such as cars, poles, etc.
- The enhanced cochlear implant will be located on the users ear of choice.
- Information will be presented to the user in the form of audio.

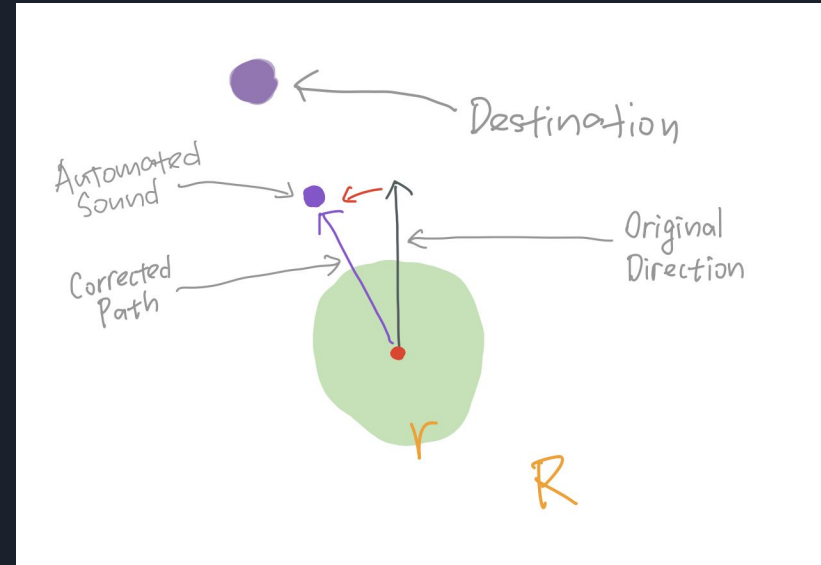


Fig. 14
Sketch 2: the radius technology

R : the bigger radius
 r : the smaller radius



How Our Device Helps Solve the Challenges

- 1). Voice guides subject to located elevator (from GPS)
- 2). Pre-loaded destination through GPS, with voice navigation
- 3). Small radius (r) helps indicate people/crowd so the user can be more aware
- 4). Large radius (R) let the user know that a car is approaching
- 5). Voice navigation provide feedback to the user, helping stay on-path
- 6). Voice and both types of radius can help give more information to the user that an intersection is approaching



Fin.



Current Literature

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- Steel, H., & Wiggett-Barnard, C. (2008). The experience of owning a guide dog. *Disability and Rehabilitation*, 30(14), 1014-1026.



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